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Showing weak stationarity

a) $X_t = (-1)^t Z_0 + (-1)^{t+1} Z_1$

$= (-1)^t Z_0 - (-1)^t Z_1$

$= (-1)^t (Z_0 - Z_1)$

Need to show μ does not depend on t

$E[X_t] = (-1)^t (E[Z_0] - E[Z_1])$

$= (-1)^t (0 - 0)$ 0 mean by defn

$= (-1)^t (0)$

$= 0$

$\therefore$ constant μ $\forall t$ ✓

Need to show $Cov(X_t, X_{t+h}) = \gamma(h)$ does not depend on t

$\gamma(h) = Cov(X_t, X_{t+h})$ ↓ 0 from above

$= E[X_t X_{t+h}] - E[X_t]E[X_{t+h}]$

$= E[X_t X_{t+h}]$

$= E[[(-1)^t (Z_0 - Z_1)][(-1)^{t+h}(Z_0 - Z_1)]$

$= E[(-1)^{t+h}(-1)^t (Z_0 - Z_1)^2]$ $(-1)^{2t+h} = (-1)^h$ b/c $(-1)^{2t} = (-1)$ $\forall t$

$= E[(-1)^{2t+h}(Z_0^2 + Z_0 Z_1 + Z_1^2)]$ $(-1)^h$ constant in terms of t, pull out of Expec.

$= (-1)^h (E[Z_0^2] + E[Z_0]E[Z_1] + E[Z_1]^2)$ $E[Z_0 Z_1] = E[Z_0]E[Z_1]$ by indep.

$= (-1)^h (\sigma^2 + 0 + \sigma^2)$

$= 2\sigma^2 (-1)^h$

$\therefore$ no dependence on t $\forall t$ ✓

Thus, we've shown X_t is weakly stationary.

■

Calculating ACF

Above, $\gamma(h) = 2\sigma^2 (-1)^h$

$\gamma(0) = 2\sigma^2$

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \frac{2\sigma^2 (-1)^h}{2\sigma^2}$$

$\therefore \rho(h) = (-1)^h$ is the ACF for X_t.